

Quantitative Metrics for the Bilateral Ankle-Motion Experiment

Purpose

This document summarizes the quantitative metrics used to evaluate the bilateral ankle-motion experiment. The goal is to complement the qualitative plots with numerical evidence that supports the comparison between the MuJoCo simulation backend and the real-robot ROS 2 backend. The metrics were computed from the MATLAB figure files containing the reference trajectories, simulated feedback, and experimental feedback for the ankle pitch and roll motions. The analysis was performed over the interval from 3 s to 5 s, which corresponds to the active motion region shown in the figures.

Signals Used

For each ankle motion, the following signals were extracted from the MATLAB figures:

- $q_{\text{ref}}(t)$: reference joint position.
- $q_{\text{sim}}(t)$: simulated joint feedback.
- $q_{\text{exp}}(t)$: experimental joint feedback from the real robot.

The analysis was performed for two bilateral motions ankle pitch and roll. Here, “bilateral” means that the motion corresponds to the left and right ankle joints following the same type of reference motion.

Tracking Error

The tracking error measures how closely the feedback signal follows the reference signal. For simulation and experiment, the errors are defined as

$$e_{\text{sim}}(t) = q_{\text{ref}}(t) - q_{\text{sim}}(t), \quad (1)$$

$$e_{\text{exp}}(t) = q_{\text{ref}}(t) - q_{\text{exp}}(t). \quad (2)$$

A smaller tracking error means that the joint feedback follows the desired reference more accurately.

RMS Tracking Error

The RMS tracking error represents the average tracking error over time. It is computed as

$$e_{\text{RMS}} = \sqrt{\frac{1}{N} \sum_{k=1}^N e^2(k)}. \quad (3)$$

This metric was selected because it gives a compact measure of the overall tracking quality. It penalizes larger errors more strongly than a simple mean absolute error.

Peak Tracking Error

The peak tracking error represents the maximum absolute error observed during the motion:

$$e_{\text{peak}} = \max_k |e(k)|. \quad (4)$$

This metric was included because it shows the worst-case deviation from the reference. This is important in real robotic systems, where a short but large deviation may still be relevant for safety or stability.

Estimated Delay

The delay estimates the temporal shift between the reference signal and the feedback signal. It was estimated by shifting the signals and selecting the lag that minimizes the RMS difference between them. The delay is reported in milliseconds:

$$T_{\text{delay}} = k_{\text{lag}} \Delta t \cdot 1000, \quad (5)$$

where k_{lag} is the estimated lag in samples and Δt is the sampling period. This metric was selected because it provides information about response timing. In a Sim-to-Real framework, delay is important because the real robot may introduce communication, sensing, or actuation latency.

Sampling Rate

The sampling rate was estimated from the time vector extracted from the MATLAB figures:

$$f_s = \frac{1}{\text{mean}(\Delta t)}. \quad (6)$$

Both simulation and experimental signals showed a sampling rate of 500 Hz. The corresponding sampling period is

$$\Delta t = 0.002 \text{ s}. \quad (7)$$

Timing jitter was not reported because the extracted time vectors were uniformly sampled. Therefore, the timestamp variation was negligible for the available figure data.

Experimental-to-Simulation Ratio

The column Exp/Sim [%] compares the experimental value with the simulated value:

$$\text{Exp/Sim}[\%] = 100 \frac{\text{Experimental value}}{\text{Simulated value}}. \quad (8)$$

The interpretation is:

- 100% means that the experimental value is equal to the simulated value.
- Below 100% means that the experimental value is lower than the simulated value.
- Above 100% means that the experimental value is higher than the simulated value.

For tracking error and delay, lower values are better.

Quantitative Results

Table 1: Quantitative ankle-motion results.

Metric	Sim	Exp	Exp/Sim [%]
<i>Bilateral ankle pitch</i>			
RMS tracking error [rad]	0.0609	0.0372	61.2
Peak tracking error [rad]	0.5197	0.0638	12.3
Delay [ms]	30	24	80.0
Sampling rate [Hz]	500	500	100.0
<i>Bilateral ankle roll</i>			
RMS tracking error [rad]	0.0155	0.0076	48.8
Peak tracking error [rad]	0.0216	0.0156	72.0
Delay [ms]	28	12	42.9
Sampling rate [Hz]	500	500	100.0

Interpretation

The results show that the experimental backend achieved stable tracking for both bilateral ankle pitch and roll motions. For pitch, the experimental RMS tracking error was 0.0372 rad, corresponding to 61.2% of the simulated RMS error. For roll, the experimental RMS tracking error was 0.0076 rad, corresponding to 48.8% of the simulated value.

The estimated delay was also lower in the experimental execution. For pitch, the delay decreased from 30 ms in simulation to 24 ms in experiment. For roll, it decreased from 28 ms to 12 ms. Both backends operated at 500 Hz, which provides a consistent basis for comparison.

Simulation–Real Discrepancy

The direct simulation–real discrepancy was not included in the main table to keep it compact. This discrepancy compares the simulated feedback directly against the experimental feedback:

$$e_{\text{sim-real}}(t) = q_{\text{sim}}(t) - q_{\text{exp}}(t). \quad (9)$$

The computed values were:

- Pitch: RMS discrepancy = 0.0513 rad, peak discrepancy = 0.5205 rad.
- Roll: RMS discrepancy = 0.0091 rad, peak discrepancy = 0.0218 rad.

These values indicate that the simulation and experimental responses are closer for roll than for pitch. The large pitch peak discrepancy suggests the presence of a short transient or local mismatch, which should be interpreted carefully together with the RMS value.